

Production and Costs

Concept of Production

Production:

- Production is a **transformation** of **resources or inputs** such as capital, equipment, labor, and land into **output** - goods and services.
- You may think of production the way you might think of making a cake.
- It takes certain ingredients to make a cake—sugar, flour, and so on.
- Similarly, it takes certain resources, or inputs, to produce a computer, a haircut, a piece of furniture, or a house.

Factors of Production

- **Inputs** are commodities or services that are used to produce goods and services.
- **Outputs** are the various useful goods or services that result from the production process and are either consumed or employed in further production.
- **Factors of production:** These can be classified into four broad categories: land, labor, and capital and entrepreneurship.
 - 1. Land**—or, more generally, **natural resources**— represents the gift of nature to our societies.

It consists of the **land** used for farming or for underpinning houses, factories, and roads; the **energy resources** that fuel our cars and heat our homes; and the **non-energy** resources like copper and sand.
 - 2. Labor** consists of the human time spent in production—working in automobile factories, writing software, teaching school, or baking pizzas.

Factors of Production

3. *Capital* resources form the durable goods of an economy, produced in order to produce yet other goods.

- Capital goods include machines, roads, computers, software, trucks, steel mills, automobiles, washing machines, and buildings.

4. Entrepreneurship refers to the particular talent that some people have for organizing the resources of land, labor, and capital to produce goods, seek new business opportunities, and develop new ways of doing things.

Production Function

- **Production function:** The **production function** specifies the **maximum** amount of output that can be produced from any **specified** set of **inputs**, **given existing technology**.

$$Q = f(L, K, L_d)$$

where

Q = output

L = Labour

K = Capital

L_d = Land

- For **example** consider the production function for **generating electricity**.
- Visualize it as a book with technical specifications for **different kinds of plants**.
- One page is for **gas turbines**, showing their inputs (initial capital cost, fuel consumption, and the amount of labor needed to run the turbine) and their outputs (amount of electricity generated).
- The next page shows inputs and outputs of **coal-fired** generating plants.
- Yet other pages describe **nuclear power** plants, solar power stations, and so forth.
- Taken together, they constitute the production function for electricity generation.

Types of Inputs

- There are two types of inputs in the production process: fixed and variable.
- ## 1. Fixed input
- A **fixed input** is an input whose quantity cannot be changed as output changes.
 - For example, suppose the McMahon and McGee Bookshelf Company has rented a **factory** under a six-month lease: McMahon and McGee, the owners of the company, have contracted to pay the \$2,500 monthly rent for **six** months—no matter what.
 - Whether McMahon and McGee produce 1 bookshelf or 7,000 bookshelves, the \$2,500 rent for the factory must be paid.
 - The **factory** is an **input** in the production process of bookshelves; specifically, it is a fixed input.

Types of Inputs (Cont.....)

2. Variable input

- A **variable input** is an input whose quantity can be changed as output changes.
- Examples of variable inputs for the McMahon and McGee Bookshelf Company include wood, paint, and so on.
- These inputs can (and most likely will) change as the production of bookshelves changes.
- As more bookshelves are produced, more of these inputs will be purchased by McMahon and McGee;
- as fewer bookshelves are produced, fewer of these inputs will be purchased.
- **Labor** might also be a variable input for McMahon and McGee.
- As they produce more bookshelves, they might hire more employees; as they produce fewer bookshelves, they might lay off some employees

Short Run and Long Run

Short Run

- A period of time in which some inputs in the production process are fixed.
- If any of the inputs of a firm are fixed inputs, then it is said to be producing in the *short run*.
- **Example**

Long Run

- A period of time in which all inputs in the production process can be varied (no inputs are fixed).
- If none of the inputs of a firm are fixed inputs—if all inputs are variable—then the firm is said to be producing in the *long run*.
- **Example**

Total, Average, and Marginal Product

- Starting with a firm's production function, we can calculate three important production concepts: total product, marginal product and , average product.

1. Total product (or output)

- The total amount of a commodity produced, measured in physical units such as bushels of wheat, tons of steel, or number of haircuts.

2. Marginal Physical Product (*MPP*)

- The change in output that results from changing the variable input by one unit, holding all other inputs fixed.
- (The **marginal product** of an input is the extra output produced by 1 additional unit of that input while other inputs are held constant.)
- For example, if the variable input is labor, so here we are talking about the *MPP* of labor.
- Specifically, the *MPP* of labor is equal to the change in output, Q , that results from changing labor, L , by one unit, *holding all other inputs fixed*.

$$MPP = \frac{\Delta Q}{\Delta L}$$

The Law of Diminishing Returns

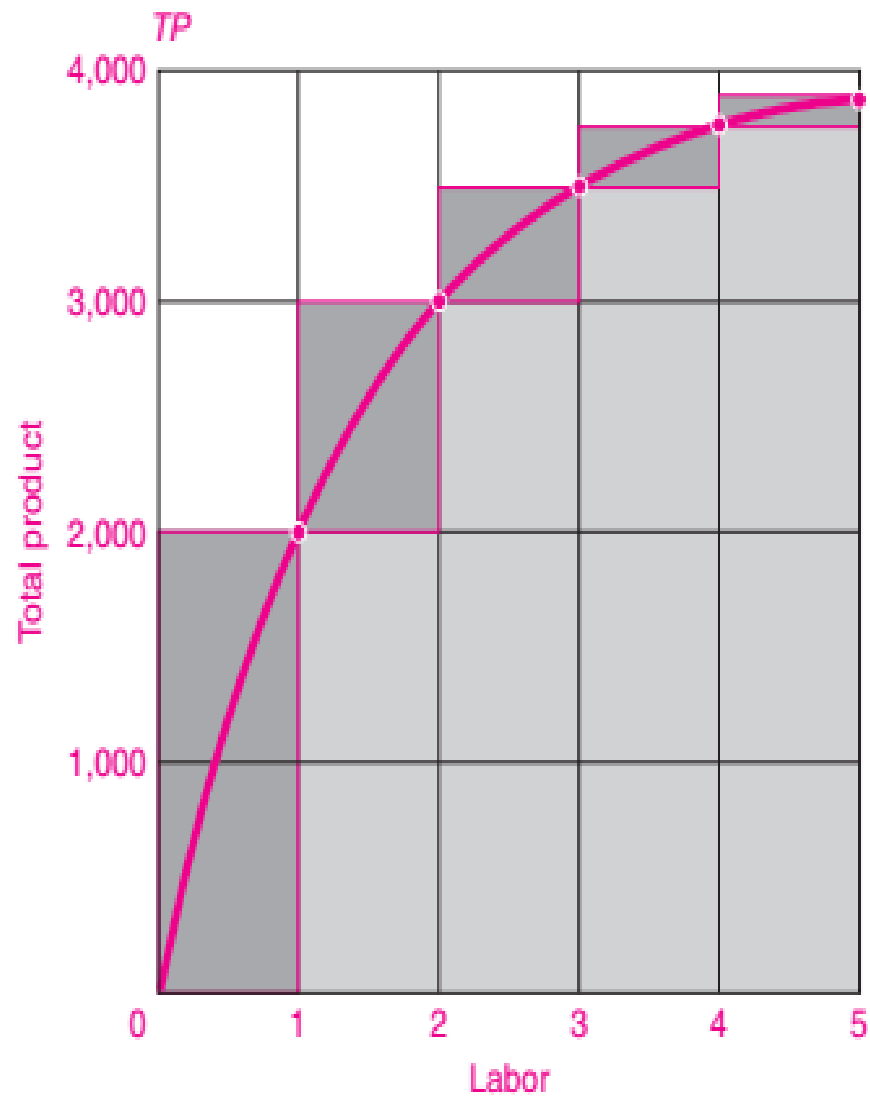
Average product

- Average product equals total output divided by total units of input.

(1) Units of labor input	(2) Total product	(3) Marginal product	(4) Average product
0	0		
1	2,000	2,000	2,000
2	3,000	1,000	1,500
3	3,500	500	1,167
4	3,800	300	950
5	3,900	100	780

TABLE 6-1. Total, Marginal, and Average Product

(a) Total Product



(b) Marginal Product

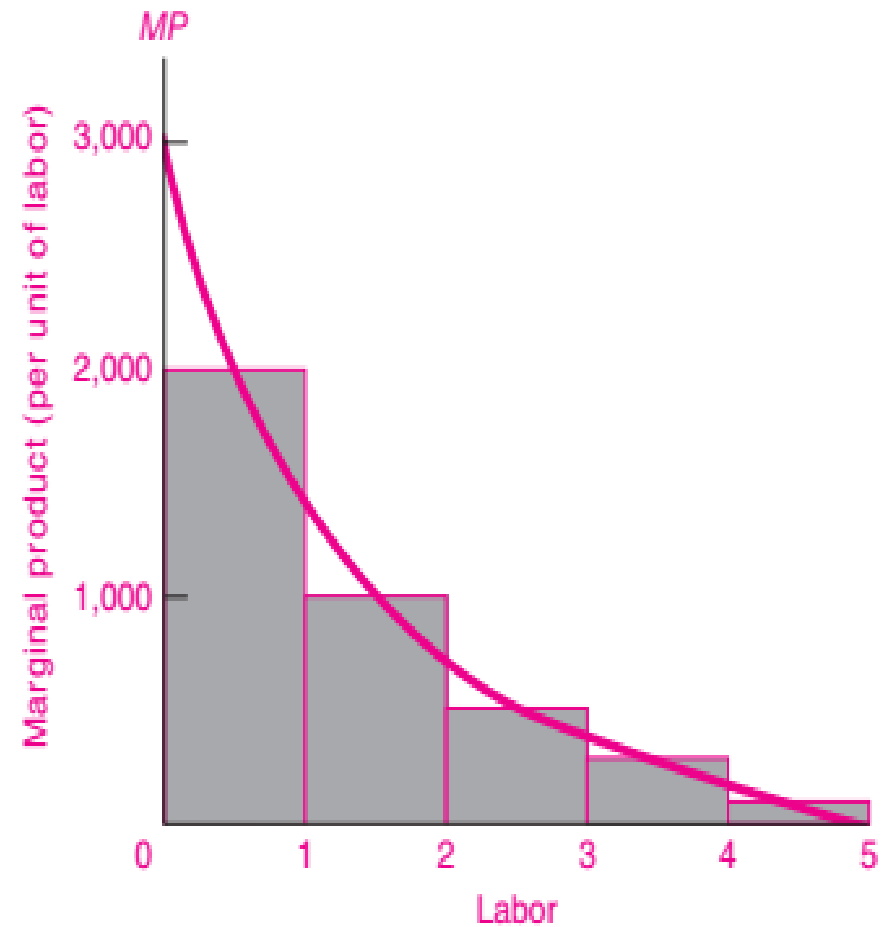


FIGURE 6-1. Marginal Product Is Derived from Total Product

The Law of Diminishing Returns (Cont.....)

(1) Fixed Input, Capital (units)	(2) Variable Input, Labor (workers)	(3) Quantity of Output, Q (units)	(4) Marginal Physical Product of Variable Input (units) $\Delta(3)/\Delta(2)$
1	0	0	
1	1	18	18
1	2	37	19
1	3	57	20
1	4	76	19
1	5	94	18
1	6	111	17
1	7	127	16
1	8	137	10
1	9	133	-4
1	10	125	-8

The Law of Diminishing Returns (Cont.....)

- Under the **law of diminishing returns**, a firm will get less and less extra output when it adds additional units of an input while holding other inputs fixed.
- In other words, the marginal product of each unit of input will decline as the amount of that input increases, holding all other inputs constant.
- As more of an input such as labor is added to a fixed amount of land, machinery, and other inputs, the labor has less and less of the other factors to work with.
- The land gets more crowded, the machinery is overworked, and the marginal product of labor declines.
- **Table 6 -1** illustrates the law of diminishing returns.
- Given fixed land and other inputs, we see that there is zero total output of corn with zero inputs of labor.
- When we add our first unit of labor to the same fixed amount of land, we observe that 2000 bushels of corn are produced.

The Law of Diminishing Returns (Cont.....)

- In our next stage, with 2 units of labor and fixed land, output goes to 3000 bushels.
- Hence, the second unit of labor adds only 1000 bushels of additional output.
- The third unit of labor has an even lower marginal product than does the second, and the fourth unit adds even less.
- **Table 6-1** thus illustrates the law of diminishing returns.
- **Figure 6-1** also illustrates the **law of diminishing returns for labor**.
- Here we see that the marginal product curve in (b) declines as labor inputs increase, which is the precise meaning of diminishing returns.
- **In Figure 6-1(a)**, diminishing returns are seen as a concave or dome-shaped total product curve.

The Law of Diminishing Returns (Cont.....)

- **What is true for labor is also true for any other input.**
- We can interchange land and labor, now holding labor constant and varying land.
- We can calculate the marginal product of each input (labor, land, machinery, water, fertilizer, etc.), and
- the marginal product would apply to any output (wheat, corn, steel, soybeans, and so forth).
- We would find that other inputs also tend to show the law of diminishing returns.
- **Diminishing returns are a key factor in explaining why many countries in Asia are so poor.**
- Living standards in crowded Rwanda or **Bangladesh** are low because there are so many workers per acre of land and not because farmers are ignorant or fail to respond to economic incentives.
- It has been found in numerous empirical studies, but exceptions have also been uncovered.
- Moreover, **diminishing returns might not hold for all levels of production.**

Laws of Returns to Scale

- **Returns to Scale** explains the behavior of output in response to a proportional and simultaneous change in input.
- When a firm increases both the inputs, there are three technical possibilities –
- **1. Constant returns to scale** denote a case where a change in all inputs leads to a proportional change in output.
- For example, if labor, land, capital, and other inputs are doubled, then under constant returns to scale output would also double. Many handicraft industries (such as haircutting in America or handloom operation in a developing country) show constant returns.

Increasing Returns to Scale

- **2. Increasing returns to scale** arise when an increase in all inputs leads to a more-than-proportional increase in the level of output.
- For example, an engineer planning a small-scale chemical plant will generally find that increasing the inputs of labor, capital, and materials by 10 percent will increase the total output by more than 10 percent.
- Engineering studies have determined that many **manufacturing processes** enjoy modestly increasing returns to scale for plants up to the largest size used today.

Decreasing Returns to Scale

- **3. Decreasing returns to scale** occur when a balanced increase of all inputs leads to a less than proportional increase in total output.
- These might arise because the **costs of management or control** become large.
- Many productive activities involving natural resources, such as providing clean drinking water to a city, show decreasing returns to scale.

Technological Change and Production

- **Technological change.**
- A **change** in the **process** of production or an introduction of a **new product** such that more or improved output can be obtained from the same bundle of inputs.
- It results in an outward shift in the production possibility curve. Often called **technological progress**.
- **Example** of technological change : improvements in computer technologies that have increased computational power by more than 1000 times in three decades.

Technological Change and Production (Cont.....)

Process innovation

- which occurs when new engineering knowledge improves production techniques for existing products.
- For example, a process innovation allows firms to produce more output with the same inputs or to produce the same output with fewer inputs.
- In other words, a process innovation is equivalent to a shift in the production function.

Product innovation

- whereby new or improved products are introduced in the marketplace.

Technological Change and Production (Cont.....)

- **Figure 6-3 illustrates** how technological change, in the form of a **process innovation**, would shift the total product curve.
- The lower line represents the feasible output, or production function, for a particular industry in the year 1995.
- Suppose that productivity, or output per unit of input, in this industry is rising at 4 percent per year.
- If we return to the same industry a decade later, we would likely see that changes in technical and engineering knowledge have led to a 48 percent improvement in output per unit of input.
- Next, consider **product innovations**, which involve new and improved products.
- It is much more difficult to quantify the importance of product innovations, but they may be even more important in **raising living standards than process innovations**.
- Many of today's goods and services did not even exist 50 years ago.

Technological Change and Production (Cont.....)

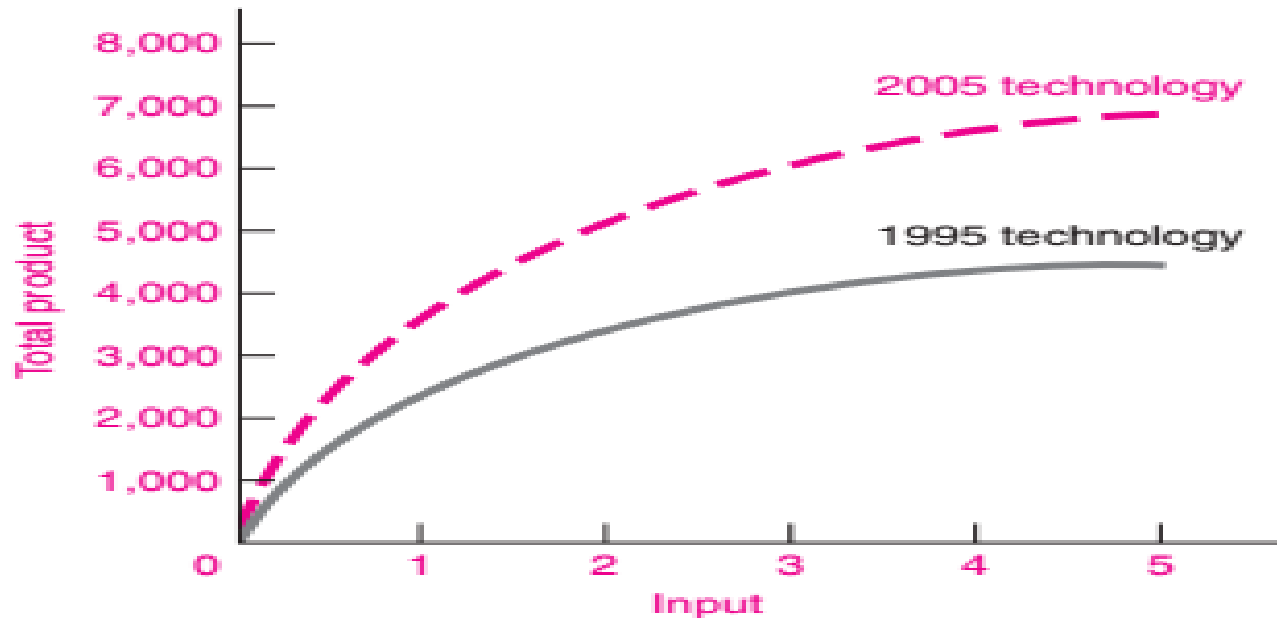


FIGURE 6-3. Technological Change Shifts Production Function Upward

Figure 6-3 shows the happy case of a technological advance. **Is the opposite case—technological regress—possible?** The answer is no for a well functioning market economy. Inferior technologies are **unprofitable** and tend to be discarded in a market economy, while more productive technologies are introduced because they increase the **profits** of the innovating firms.

Fixed, Variable and Total Costs

- **Fixed Costs (FC):**

Fixed costs are expenses that must be paid even if the firm produces zero output. Sometimes it is called “**overhead**” or “**sunk costs**.”

- Costs that do not vary with output; the costs associated with fixed inputs.
- **For example**, a law firm might have an office lease which runs for 10 years and remains an obligation even if the firm shrinks to half its previous size.
- Rent for factory or office space, interest payments on debts, and so forth.

- **Variable Costs (VC)**

Variable costs do vary as output changes.

- Costs that vary with output; the costs associated with variable inputs.
- **Examples** include materials required to produce output (such as steel to produce automobiles), production workers wages, power to operate factories, and so on.

Fixed, Variable and Total Costs

- **Total Cost (TC)**

The sum of fixed costs and variable costs.

$$TC = TFC + TVC$$

(1) Quantity q	(2) Fixed cost FC (\$)	(3) Variable cost VC (\$)	(4) Total cost TC (\$)
0	55	0	55
1	55	30	85
2	55	55	110
3	55	75	130
4	55	105	160
5	55	155	210
6	55	225	280

TABLE 7-1. Fixed, Variable, and Total Costs

Fixed, Variable and Total Costs

- **Total cost** represents the lowest total dollar expense needed to produce each level of output q . TC rises as q rises.
- **Fixed cost** represents the total dollar expense that is paid out even when no output is produced; fixed cost is unaffected by any variation in the quantity of output.
- **Variable cost** represents expenses that vary with the level of output—such as raw materials, wages, and fuel—and includes all costs that are not fixed.
- Always, by definition,

$$TC = TFC + TVC$$

Marginal Cost

- **Marginal Cost (*MC*)**

Marginal cost (*MC*) denotes the extra or additional cost of producing 1 extra unit of output.

- Say a firm is producing 1000 compact discs for a total cost of \$10,000. If the total cost of producing 1001 discs is \$10,006, then the marginal cost of production is \$6 for the 1001st disc.
- The change in total cost that results from a change in output:

$$MC = \Delta TC / \Delta Q$$

(1) Output <i>q</i>	(2) Total cost <i>TC</i> (\$)	(3) Marginal cost <i>MC</i> (\$)
0	55	
1	85	30
2	110	25
3	130	20
4	160	30
5	210	50

TABLE 7-2. Calculation of Marginal Cost

The Relationship between Total Costs and Marginal Cost

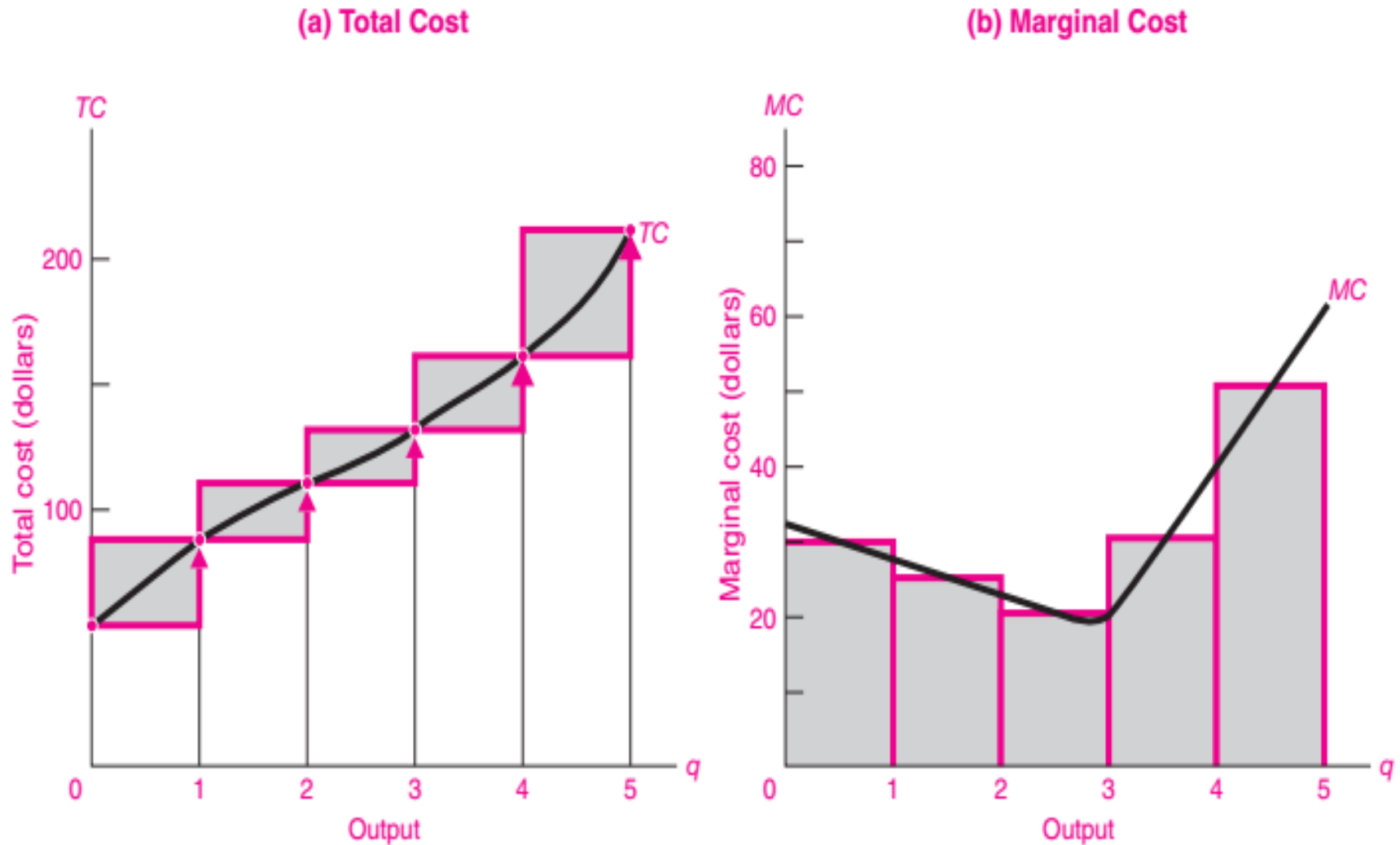


FIGURE 7-1. The Relationship between Total Cost and Marginal Cost

Average Fixed Cost, Average Variable Cost, Average Total Cost

- **Average cost** is the total cost divided by the total number of units produced.
- That is,

$$\text{Average cost} = \frac{\text{total cost}}{\text{output}} = \frac{TC}{q} = AC$$

Average Fixed Cost (AFC)

Total fixed cost divided by quantity of output: $AFC = TFC / Q$.

Average Variable Cost (AVC)

Total variable cost divided by quantity of output: $AVC = TVC / Q$.

Average Total Cost (ATC)

Total cost divided by quantity of output: $ATC = TC / Q$

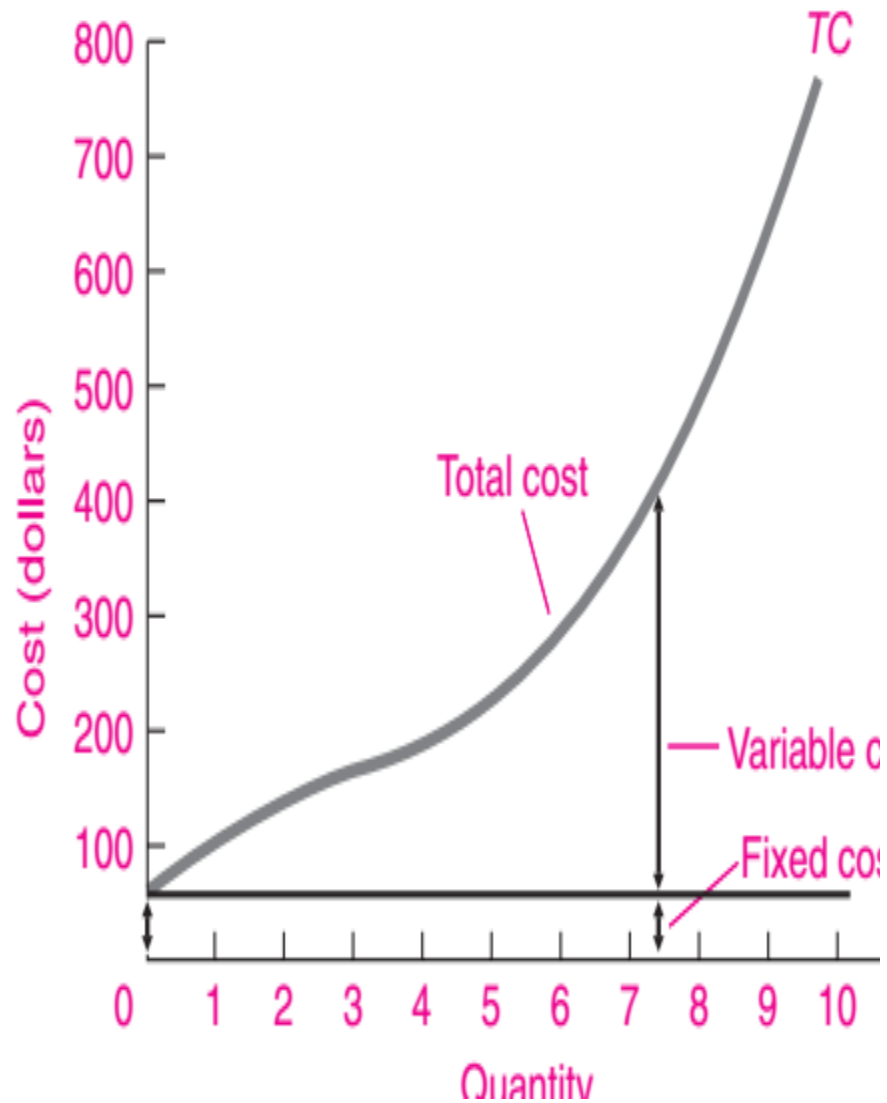
All Cost Concepts

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Quantity q	Fixed cost FC (\$)	Variable cost VC (\$)	Total cost $TC = FC + VC$ (\$)	Marginal cost per unit MC (\$)	Average cost per unit $AC = TC/q$ (\$)	Average fixed cost per unit $AFC = FC/q$ (\$)	Average variable cost per unit $AVC = VC/q$ (\$)
0	55	0	55		Infinity	Infinity	Undefined
1	55	—	85	30	85	55	30
2	—	55	110	25	55	—	27½
3	55	75	130	—	43⅓	18⅓	25
4*	55	105	160	30 40*	40*	13¾	26¼
5	55	155	210	50	42	11	—
6	55	225	280	70	46⅔	9%	37½

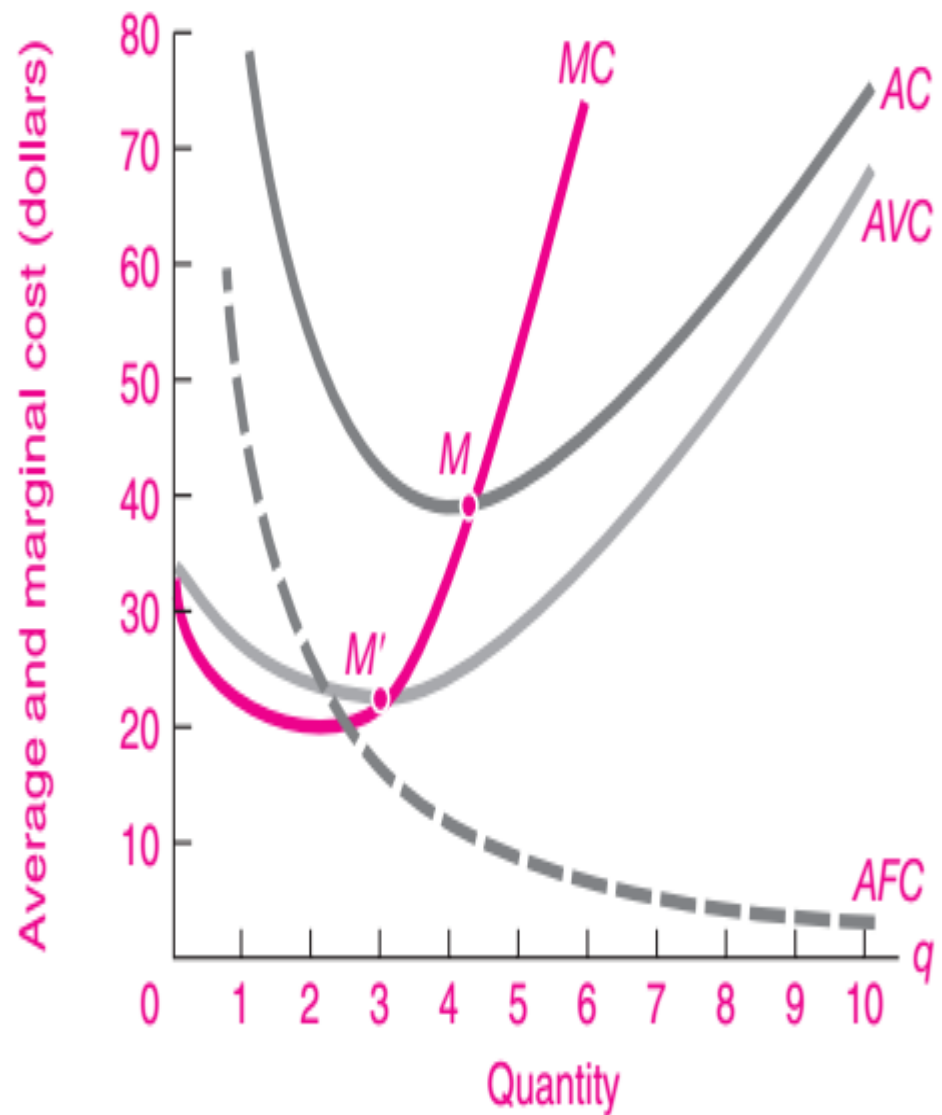
*Minimum level of average cost.

TABLE 7-3. All Cost Concepts Derive from Total Cost Schedule

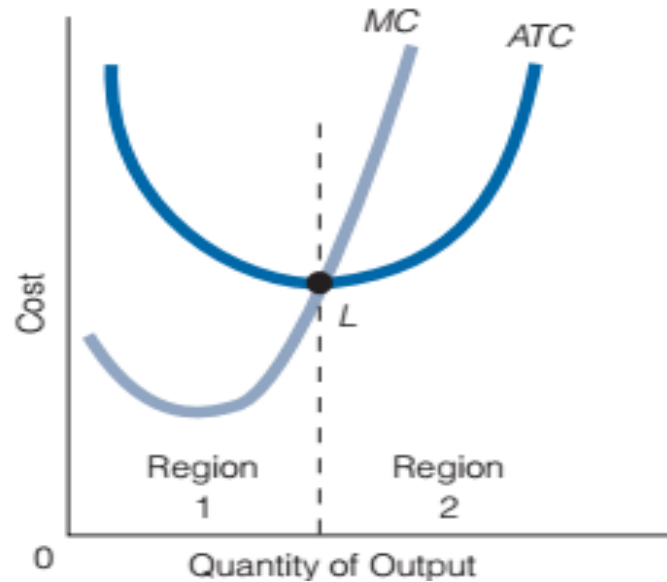
(a) Total, Fixed, and Variable Cost



(b) Average Cost, Marginal Cost



The Relation between Average Cost and Marginal Cost



1. When marginal cost is below average cost, it is pulling average cost down.
2. When MC is above AC , it is pulling up AC .
3. When MC just equals AC , AC is constant. At the bottom of a U-shaped AC , $MC = AC$ minimum = AC .

long Run Cost and Short Run Cost

- **long run cost:** The **long run** is a **period** of time in which all factors of production and costs **are** variable. In the **long run**, firms **are** able to adjust all costs.
- **Short run cost:** There are both fixed and variable factors in the **short run**. In the **short run**, firms **are** only able to influence prices through adjustments made to production levels.

The Link Between Production and Costs

- **What are the factors that determine the cost curves?**
- The key elements are
 - (1) factor prices and
 - (2) the firm's production function.
- Clearly the prices of inputs like labor and land are important ingredients of costs.
- Higher rents and higher wages mean higher costs
- But costs also depend on the firm's technological opportunities.
- If technological improvements allow the firm to produce the same output with fewer inputs, the firm's costs will fall.

The Link Between Production and Costs (Cont.....)

- Indeed, if you know factor prices and the production function, you can calculate the cost curve.
- We can show the derivation of cost from production data and factor prices in the numerical example shown in Table 7-5.
- Suppose Farmer Smith rents 10 acres of land and can hire farm labor to produce wheat.
- Per period, land costs \$5.5 per acre and labor costs \$5 per worker.
- Using up-to-date farming methods, Smith can produce according to the production function shown in the first three columns of Table 7-5.
- In this example, land is a fixed cost (because Farmer Smith operates under a 10-year lease), while labor is a variable cost (because farmworkers can easily be hired and fired).

The Link Between Production and Costs (Cont.....)

- Using the production data and the input-cost data, for each level of output we calculate the total cost of production shown in column (6) of Table 7-5.
- As an example, consider the total cost of production for 3 tons of wheat.
- Using the given production function, Smith can produce this quantity with 10 acres of land and 15 farmhands.
- The total cost of producing 3 tons of wheat is $(10 \text{ acres} \times \$5.5 \text{ per acre}) + (15 \text{ workers} \times \$5 \text{ per worker}) = \130 .
- Similar calculations will give all the other total cost figures in column (6) of Table 7-5.
- Note that these total costs are identical to the ones shown in Tables 7-1 through 7-3, so the other cost concepts shown in the tables (i.e., *MC*, *FC*, *VC*, *AC*, *AFC*, and *AVC*) are also applicable to the production-cost example of Farmer Smith.

TABLE 7-5. Costs are Derived from Production Data and Input Costs

(1) Output (tons of wheat)	(2) Land inputs (acres)	(3) Labor inputs (workers)	(4) Land rent (\$ per acre)	(5) Labor wage (\$ per worker)	(6) Total cost (\$)
0	10	0	5.5	5	55
1	10	6	5.5	5	85
2	10	11	5.5	5	110
3	10	15	5.5	5	130
4	10	21	5.5	5	160
5	10	31	5.5	5	210
6	10	45	5.5	5	280

Diminishing Returns and U-Shaped Cost Curves

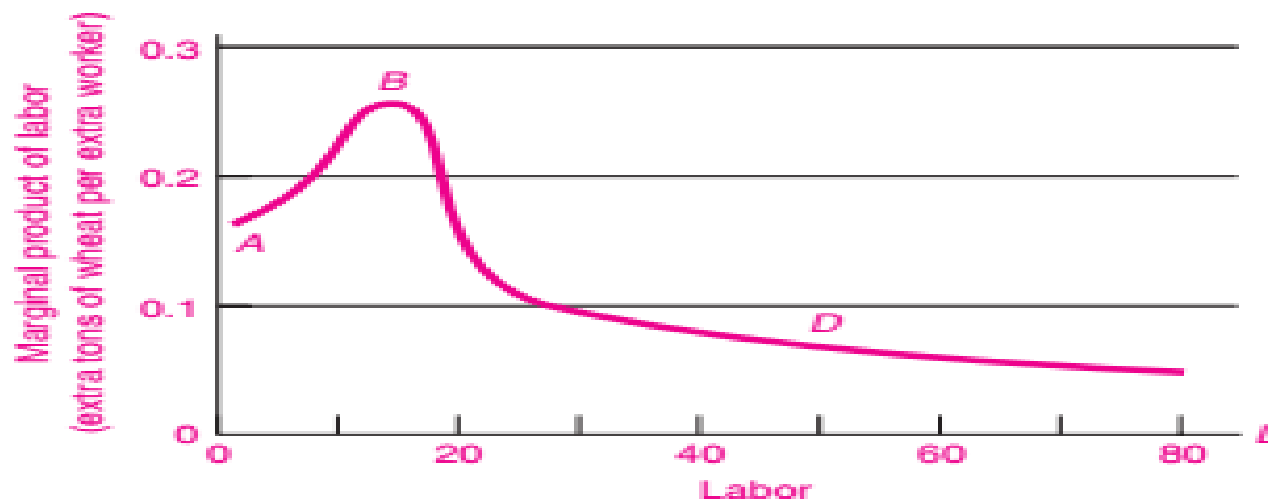
- **Why is the cost curve U-shaped?**
- For a U-shaped cost curve, cost falls in the initial phase, reaches a minimum point, and finally begins to rise.
- Let's explore the reasons.
- The analysis of cost distinguished two different time periods, the short run and the long run.
- The **short run** is the period of time that is long enough to adjust variable inputs, such as materials and production labor, but too short to allow all inputs to be changed.
- In the **short run**, fixed or overhead factors such as plant and equipment cannot be fully modified or adjusted.
- Therefore, in the short run, labor and materials costs are typically variable costs, while capital costs are fixed.
- In the **long run**, all inputs can be adjusted— including labor, materials, and capital. Hence, in the long run, all costs are variable and none are fixed.

Diminishing Returns and U-Shaped Cost Curves (Cont.....)

- Typically, in the short run, we will consider capital to be the fixed cost and labor to be the variable cost.
- That is not always true, but generally labor inputs can be adjusted more easily than can capital.
- **Why is the cost curve U-shaped?** Consider the short run in which capital is fixed but labor is variable.
- In such a situation, there are **diminishing returns** to the **variable factor** (labor) because each **additional unit of labor has less capital to work with**.
- As a result, the marginal cost of output will rise because the extra output produced by each extra labor unit is going down.
- In other words, **diminishing returns to the variable factor will imply an increasing short-run marginal cost**.
- This shows why diminishing returns lead to rising marginal costs.

Diminishing Returns and U-Shaped Cost Curves (Cont.....)

(a) Diminishing Returns . . .



(b) . . . Produce Upward-Sloping MC

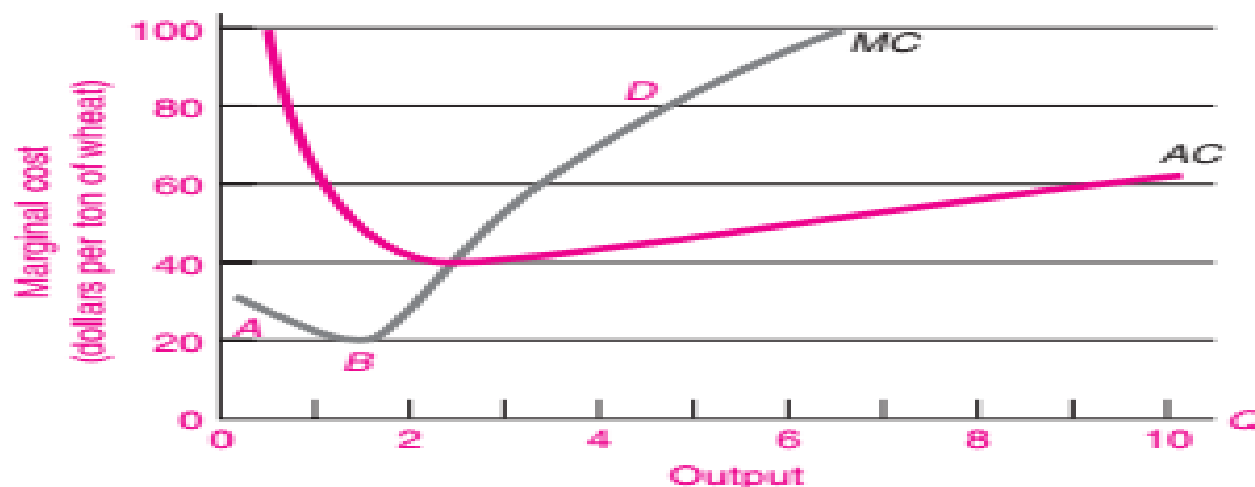


FIGURE 7-3. Diminishing Returns and U-Shaped Cost Curves

Diminishing Returns and U-Shaped Cost Curves (Cont.....)

- Figure 7-3, which contains exactly the same data as Table 7-5, illustrates the point.
- It shows that the **region of increasing marginal product corresponds to falling marginal costs, while the region of diminishing returns implies rising marginal costs.**
- We can summarize the relationship between the productivity laws and the cost curves as follows:
- In the short run, when factors such as capital are fixed, variable factors tend to show an initial phase of increasing marginal product followed by diminishing marginal product.
- The corresponding cost curves show an initial phase of declining marginal costs, followed by increasing *MC* after diminishing returns have set in.

Production and Costs in the Long Run

- In the long run, there are no fixed inputs and no fixed costs.

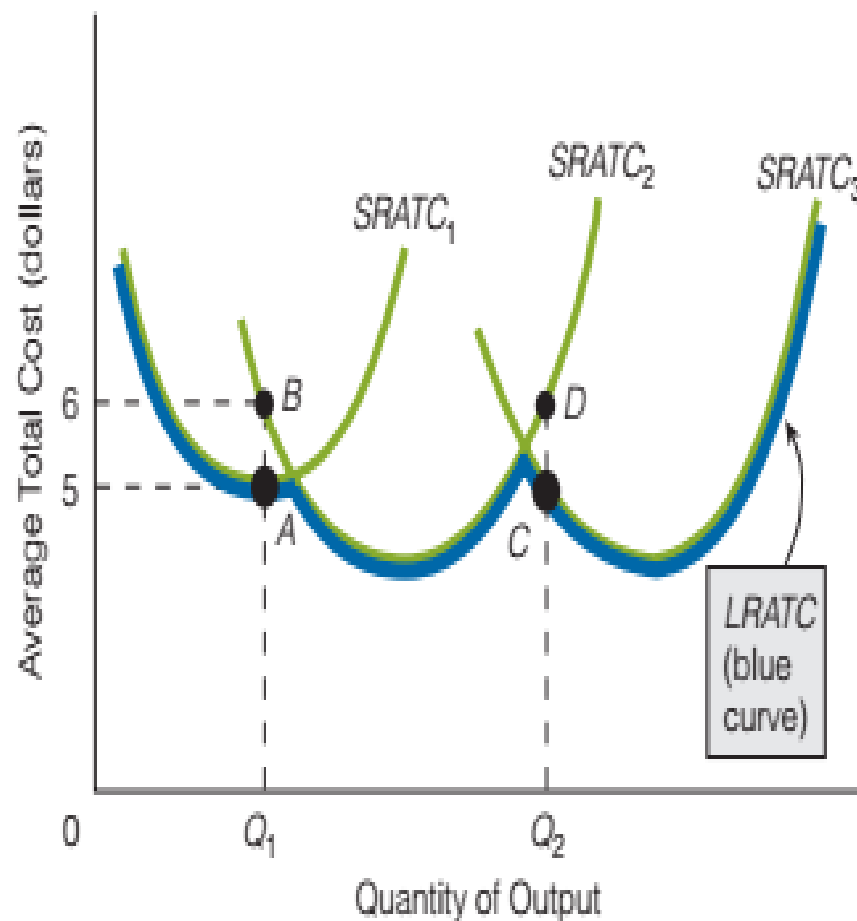
Long-Run Average Total Cost Curve

- In the long run, there are no fixed costs, so variable costs *are* total costs.
- This section focuses on
 - (1) what the long-run average total cost (*LRATC*) curve is and
 - (2) what it looks like.
- **The LRATC curve** shows the lowest unit cost at which the firm can produce any given level of output.
- Consider the manager of a firm that produces bedroom furniture.
- When all inputs are variable, the manager must decide what the situation of the firm should be in the (upcoming) short-run period.
- For example, suppose he needs to determine the size of the **plant**; that is, he must decide whether the plant will be small, medium, or large.
- After this decision is made, he is locked into a specific plant size; he is locked in for the short run.

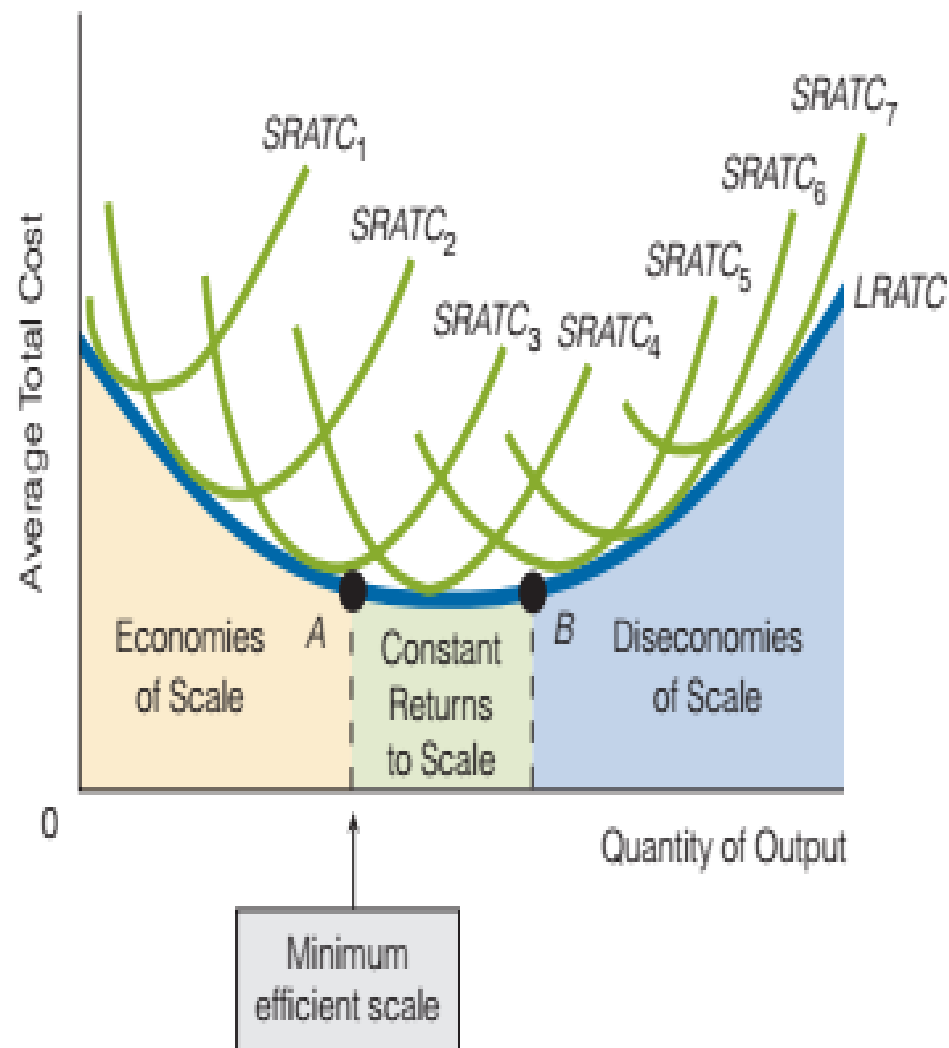
Production and Costs in the Long Run (Cont.....)

- Associated with each of the **three different plant sizes** is a short run average total cost (*SRATC*) curve.
- The three short-run average total cost curves, representing the different plant sizes, are illustrated in Exhibit 7(a).
- Suppose the manager of the firm wants to produce output level Q_1 . Which plant size will he choose?
- Obviously, he will choose the plant size represented by *SRATC*₁ because this gives a **lower unit cost** of producing Q_1 than the plant size represented by *SRATC*₂.
- The latter plant size has a higher unit cost of producing Q_1 (\$6 as opposed to \$5).
- Suppose, though, the manager chooses to produce Q_2 . Which plant size will he choose now?
- He will choose the plant size represented by *SRATC*₃ because the unit cost of producing Q_2 is lower with the plant size represented by *SRATC*₃ than it is with the plant size represented by *SRATC*₂.

Exhibit 7: Long-Run Average Total Cost Curve (LRATC)



(a)



(b)

Production and Costs in the Long Run (Cont.....)

- If we were to ask the same question for **every (possible) output level**, we would derive the **long-run average total cost (*LRATC*)** curve.
- **The *LRATC* curve shows the lowest unit cost at which the firm can produce any given level of output.**
- In Exhibit 7(a), it is those portions of the three *SRATC* curves that are tangential to the blue curve.
- The *LRATC* curve is the scalloped blue curve. Exhibit 7(b) shows a host of *SRATC* curves and one *LRATC* curve.
- The *LRATC* curve is smooth in part (b) because we assume there are many plant sizes in addition to the three represented in (a).
- In other words, although they have not been drawn, short-run average total cost curves representing different plant sizes exist in (b) between *SRATC*1 and *SRATC*2 and between *SRATC* 2 and *SRATC*3 and so on.
- In this case, the *LRATC* curve is smooth and touches each *SRATC* curve at one point.

Production and Costs in the Long Run (Cont.....)

- **Economies of Scale, Diseconomies of Scale, and Constant Returns to Scale**
- The three conditions can easily be seen in the *LRATC* curve in Exhibit 7(b).
- If economies of scale are present, the *LRATC* curve is falling, if constant returns to scale are present, the curve is flat, and if diseconomies of scale are present, the curve is rising.
- Economies of scale  *LRATC* is falling
Constant returns to scale  *LRATC* is constant
Diseconomies of scale  *LRATC* is rising
- If, in the production of a good, economies of scale give way to constant returns to scale or diseconomies of scale, as in Exhibit 7(b), the point at which this occurs is referred to as the minimum efficient scale.
- The **minimum efficient scale** is the lowest output level at which average total costs are minimized. Point A represents the minimum efficient scale in Exhibit 7(b).
- **Because no input is fixed, economies of scale, diseconomies of scale, and constant returns to scale must be relevant only in the long run.**

Production and Costs in the Long Run (Cont.....)

- Why Economies of Scale?
- Up to a certain point, long-run unit costs of production fall as a firm grows.
- There are two main reasons for this:
 - (1) Growing firms offer greater opportunities for employees to specialize. Individual workers can become highly proficient at more narrowly defined tasks, often producing more output at lower unit costs.
 - (2) Growing firms (especially large, growing firms) can take advantage of highly efficient mass production techniques and equipment that ordinarily require large setup costs and thus are economical only if they can be spread over a large number of units. For example, assembly line techniques are usually “cheap” when millions of units of a good are produced but are “expensive” when only a few thousand units are produced.

Production and Costs in the Long Run (Cont.....)

- **Why Diseconomies of Scale?**
- Diseconomies of scale usually arise at the point where **a firm's size causes coordination, communication, and monitoring problems.**
- In very large firms, managers often find it difficult to coordinate work activities, communicate their directives to the right persons in satisfactory time, and monitor personnel effectively.
- The business operation simply gets “too big.”
- There is, of course, a monetary incentive not to pass the point of operation where diseconomies of scale exist.
- Firms will usually find ways to avoid diseconomies of scale.
- They will reorganize, divide operations, hire new managers, and so on.

Production and Costs in the Long Run (Cont.....)

- Minimum Efficient Scale and Number of Firms in an Industry
- Some industries are composed of a smaller number of firms than other industries are. Or
- we can say there is a different degree of concentration in different industries.
- **Shifts in Cost Curves**
 - Taxes
 - Input Prices
 - Technology
- **Why is it the short-run average cost (SRAC) curve and the long-run average cost (LAC) curve are U-shaped? The reasons for their shape are quite different.**

Why is it the short-run average cost (SRAC) curve and the long-run average cost (LAC) curve are U-shaped?

- **Short-run** - a period of time where at least one factor of production is fixed (land, labour, capital, enterprise)
- **Long-run** - when all the factors of production are variable.
- ***The difference is:***
- In the short run, the average cost curve is u-shaped due to the **fixed costs of production being spread amongst more units produces** (the downfall), until the minimum point, where after the minimum point extra production leads to **diminishing returns**, or in other words more production may lead to the company employing an extra employee, with no additional capital which leads to him bothering the others, decreasing the overall efficiency and thus raising the costs per unit.
- In the long run however, the curve is u-shaped due to **economies of scale** like purchasing economies which decrease the cost per unit of production, until the minimum efficient scale, where afterwards, with the increase in production **diseconomies of scale set in** like communication problems, which increase the cost per unit.
- Not so different. Short-run average cost is u shaped because of diminishing marginal returns of labor with constant supply of capital. Long-run average cost reflects costs with variable capital, but because of economies and diseconomies of scale, you also see u shaped cost.

Isoquants (Equal-product curve)

- **Isoquants (Equal-product curve):** An **isoquant** is a curve that shows all the combinations of inputs that yield the same level of output. 'Iso' means equal and 'quant' means quantity. Therefore, an **isoquant** represents a constant quantity of output.

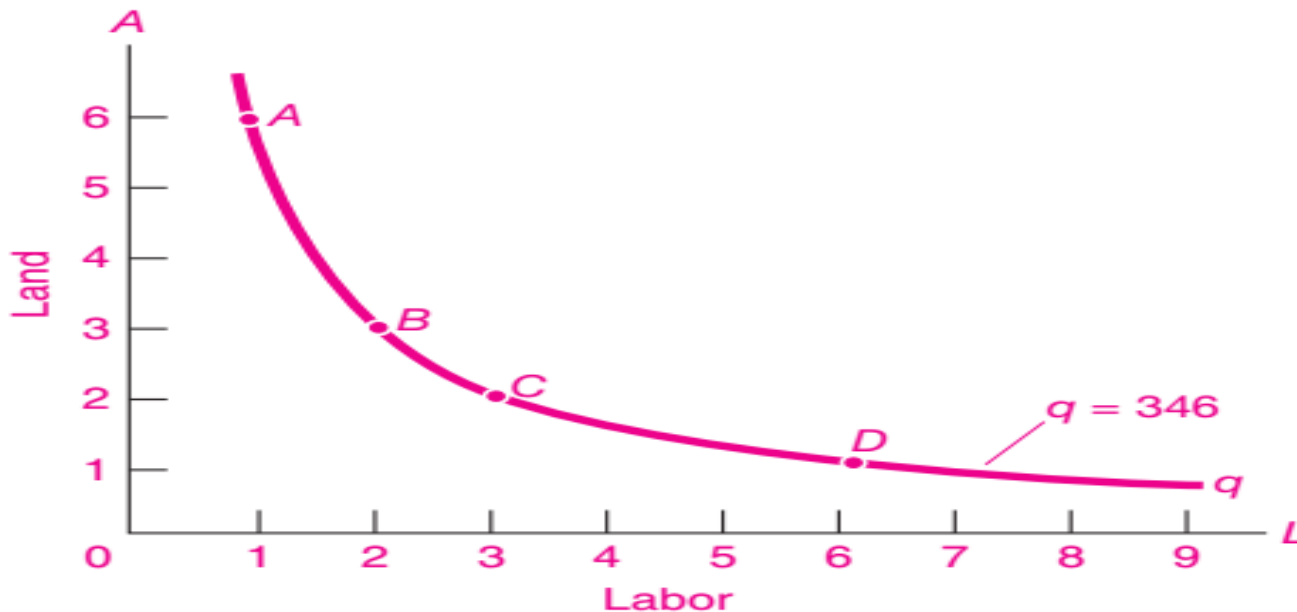


FIGURE 7A-1. Equal-Product Curve

All the points on the equal-product curve represent the different combinations of land and labor that can be used to produce the same 346 units of output.

Properties of Isoquants

- Isoquants have a negative slope.
- Isoquants are convex to the origin.
- Isoquants cannot intersect or be tangent to each other.
- Upper Isoquants represents higher level of output

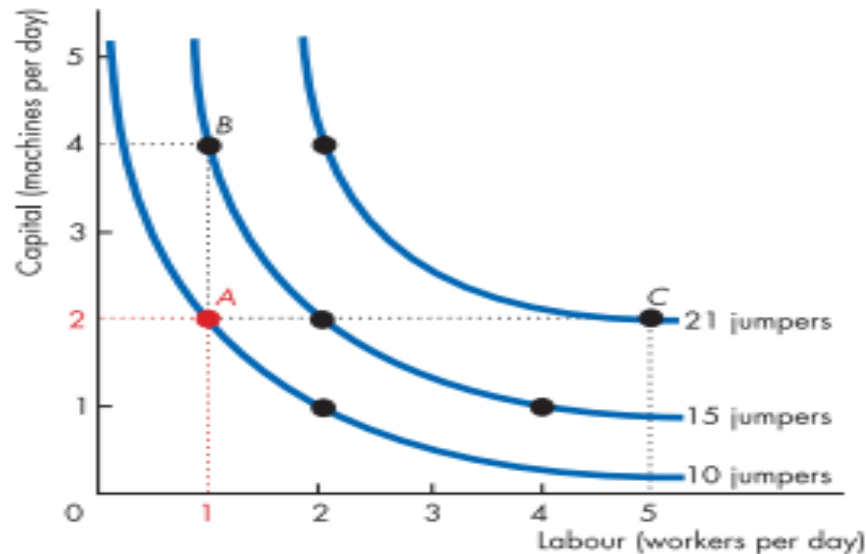
Isoquant Map

- **An Isoquant Map**

An **isoquant** is a curve that shows the different combinations of labour and capital required to produce a given quantity of output. The word isoquant means “equal quantity”. There is an isoquant for each output level. A series of isoquants is called an **isoquant map**

An Isoquant Map

1



The slope of the isoquant curve/ $MRTS_{L,K}$ /Substitution ratio

- The slope of the isoquant curve is the marginal rate of technical substitution (MRTS).
- **Marginal rate of technical substitution (labor for capital):**
measures the amount of K the firm the firm could give up in exchange for an additional L, in order to just be able to produce the same output as before.

Marginal products and the MRTS are related:

$$MP_L/MP_K = MRTS_{L,K}$$

Equal-cost (Iso-cost) line

- Equal-cost (Iso-cost) line:** A line in a graph showing the various possible combinations of factor inputs that can be purchased with a given quantity of m

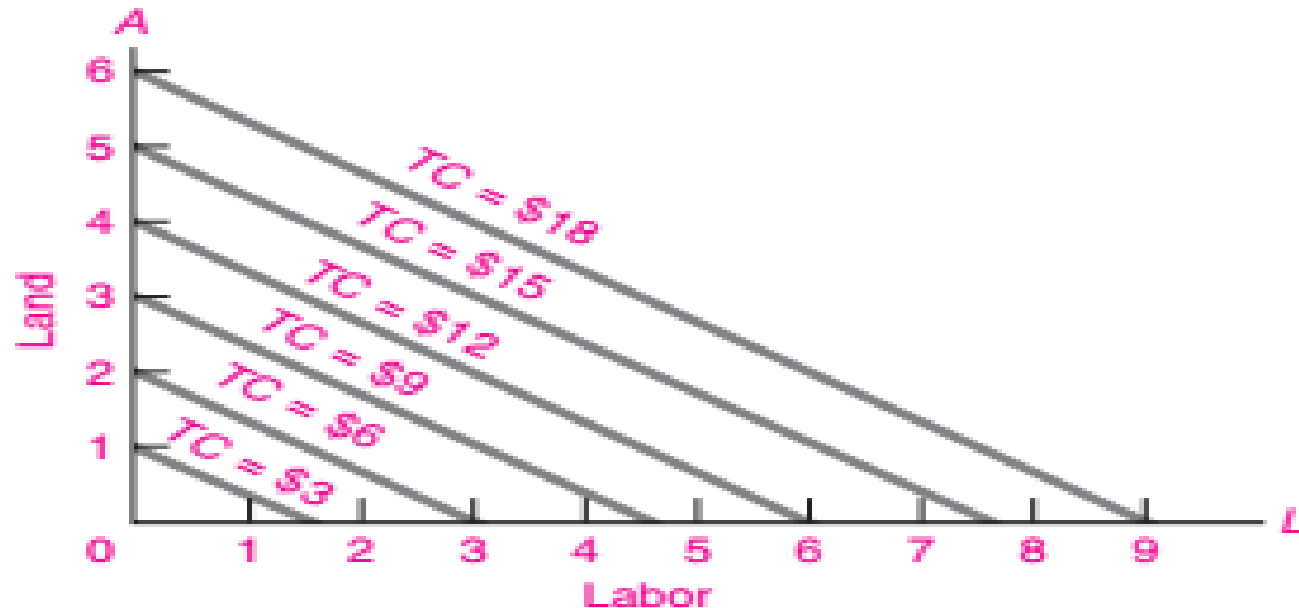


FIGURE 7A-2. Equal-Cost Lines

Every point on a given equal-cost line represents the same total cost. The lines are straight because factor prices are constant, and they all have a negative slope equal to the ratio of labor price to land price, $\$2/\3 , and hence are parallel.

Producer's Equilibrium

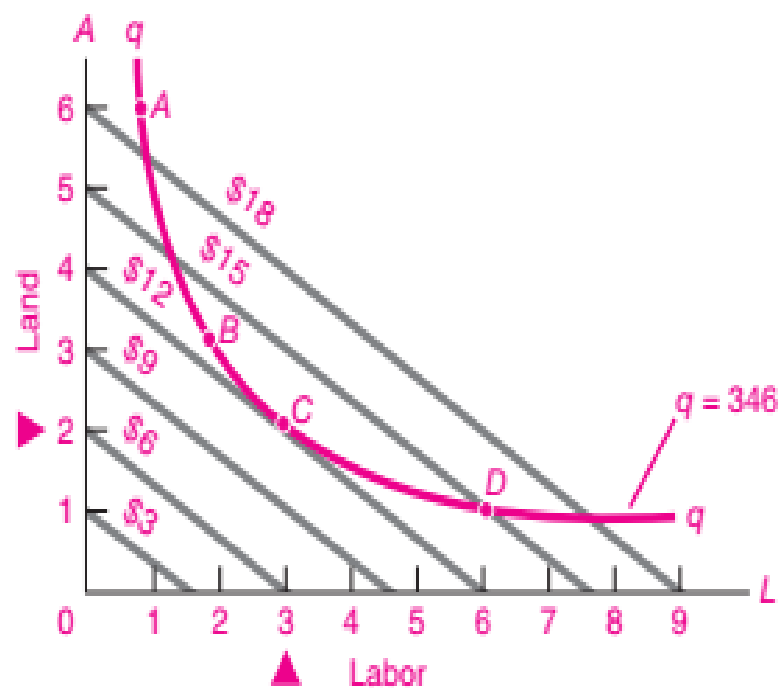


FIGURE 7A-3. Least-Cost Input Combination Comes at C

Least Cost Condition

Least-Cost Conditions

Using our graphical apparatus, we have therefore derived the conditions under which a firm will minimize its costs of production:

1. The ratio of marginal products of any two inputs must equal the ratio of their factor prices:

$$\begin{aligned}\text{Substitution ratio} &= \frac{\text{marginal product of labor}}{\text{marginal product of land}} \\ &= \frac{\text{slope of}}{\text{equal-product}} = \frac{\text{price of labor}}{\text{price of land}} \\ &\quad \text{curve}\end{aligned}$$

2. We can also rewrite condition 1 in a different and illuminating way. From the last equation it follows that the marginal product per dollar received from the (last) dollar of expenditure must be the same for every productive input:

$$\begin{aligned}\frac{\text{Marginal product of } L}{\text{Price of } L} &= \\ \frac{\text{marginal product of } A}{\text{price of } A} &= \dots\end{aligned}$$

Books Recommended

- Samuelson, P., & Nordhaus, W. (2005). Economics (18th ed.). Boston: McGraw-Hill/Irwin.
- Roger A. Arnold, Economics, 9th Edition, **Thomson Higher Education, 2009.**